



Real-Time Ferry Ticket Sales & Redemption Analytics

Toronto Island Park — An Empirical Study of Temporal Demand Patterns in Public Ferry Ticketing Systems

Prepared using Python (Pandas, NumPy, Matplotlib, Seaborn) and MySQL · Dataset: 261,538 transaction records · July 2026

EXPLORATORY DATA ANALYSIS

TIME SERIES

DEMAND FORECASTING

Abstract & Study Objectives

This study presents a comprehensive EDA of the Toronto Island Ferry Tickets dataset — 261,538 time-stamped records capturing ticket sales and redemptions. Data was ingested, cleaned, enriched with derived temporal features, and persisted to MySQL before systematic analysis. Key findings: demand peaks in the afternoon (12–3 PM), weekends dominate volume, Summer is the peak season, and sales vs. redemptions differ by only **0.8 percentage points** (50.4% vs. 49.6%), indicating operational efficiency.

01	02	03
Hourly Variation	Daily Patterns	Seasonal Trends
Quantify peak and off-peak time windows across the 24-hour day.	Identify which days of the week generate the highest ticket activity.	Analyze demand variation across Winter, Spring, Summer, and Fall.
04	05	
Sales vs. Redemptions	Rolling Averages	
Assess the gap between tickets sold and tickets actually redeemed.	Smooth short-term fluctuations to surface the underlying demand trend.	

Dataset Description & Descriptive Statistics

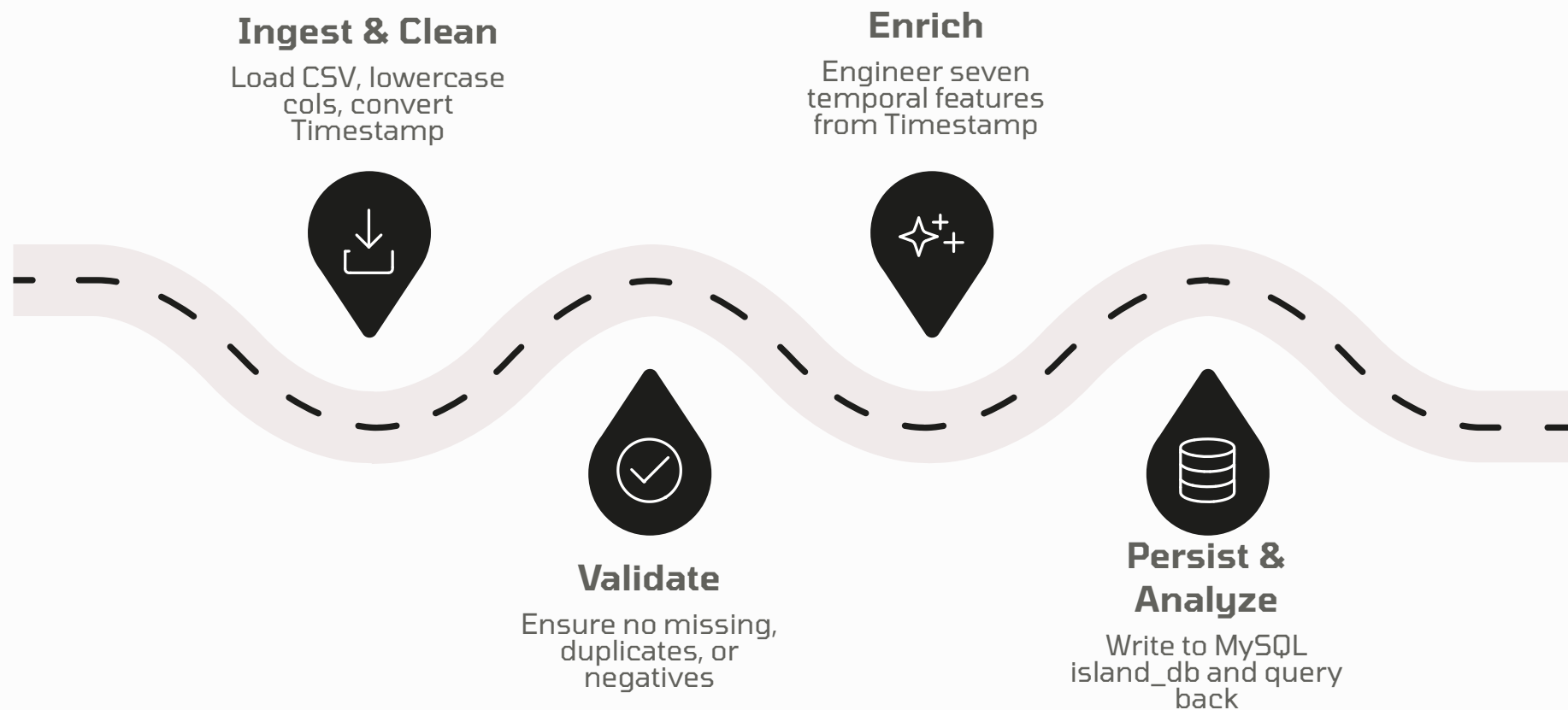
The source CSV contains **261,538 rows** and 4 columns, each representing a 15-minute interval snapshot of ticketing activity. Both sales and redemption counts exhibit a **right-skewed distribution** — the mean substantially exceeds the median, indicating that most intervals record low activity while peak hours record very high volumes.

Column	Description	Data Type
id	Unique transaction identifier	int64
Timestamp	Date/time of 15-min interval	datetime64
redemption_count	Tickets redeemed in interval	int64
sales_count	Tickets sold in interval	int64

Statistic	Redemption Count	Sales Count
Count	261,538	261,538
Mean	48.89	49.60
Std. Deviation	104.55	99.86
Median (50%)	11	13
Maximum	7,216	7,229

Methodology: Data Pipeline

The analysis followed a standard EDA pipeline executed in Python Jupyter Notebooks, decoupling data preparation from the analytical stage via MySQL persistence — mirroring a realistic production analytics workflow.



Tools & Libraries

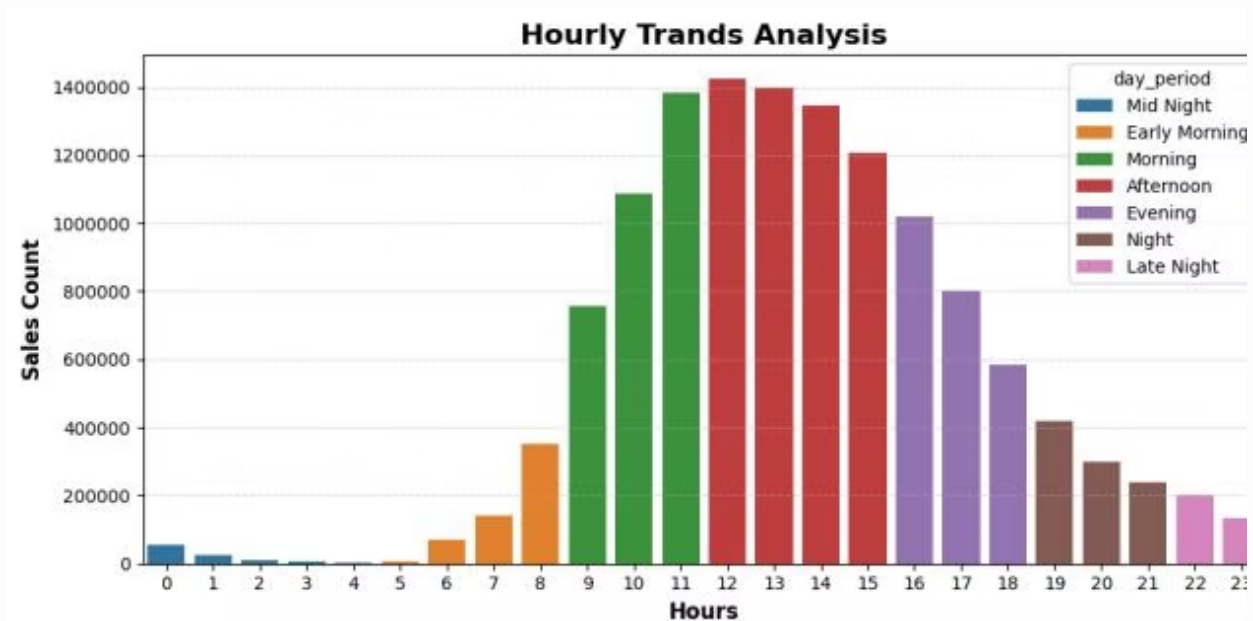
- Pandas & NumPy — data manipulation
- Matplotlib — pie charts, rolling-average plots
- Seaborn — grouped bar charts
- SQLAlchemy & mysql-connector — MySQL persistence

Engineered Features (7)

- **hours** — hour of day (0–23)
- **day_name** — full weekday name
- **month_name / month** — month info
- **season** — meteorological (Winter/Spring/Summer/Fall)
- **week_part** — Weekday vs. Weekend
- **day_period** — 6-bucket period (e.g., Afternoon)
- **gap** — sales_count minus redemption_count

Hourly Demand Trends

Ticket sales were aggregated by hour of day and day-period bucket. The afternoon window (12:00–3:00 PM) dominates, peaking at hour 12 with approximately **1.4 million** in sales — over 13× the off-peak midnight hours. Morning hours (9–11 AM) form a secondary ramp-up, while Late Night and Mid Night are nearly inactive.



🌙 Off-Peak

Mid Night (0–4) & Late Night (22–23):
~101K & ~336K total sales

🌅 Moderate

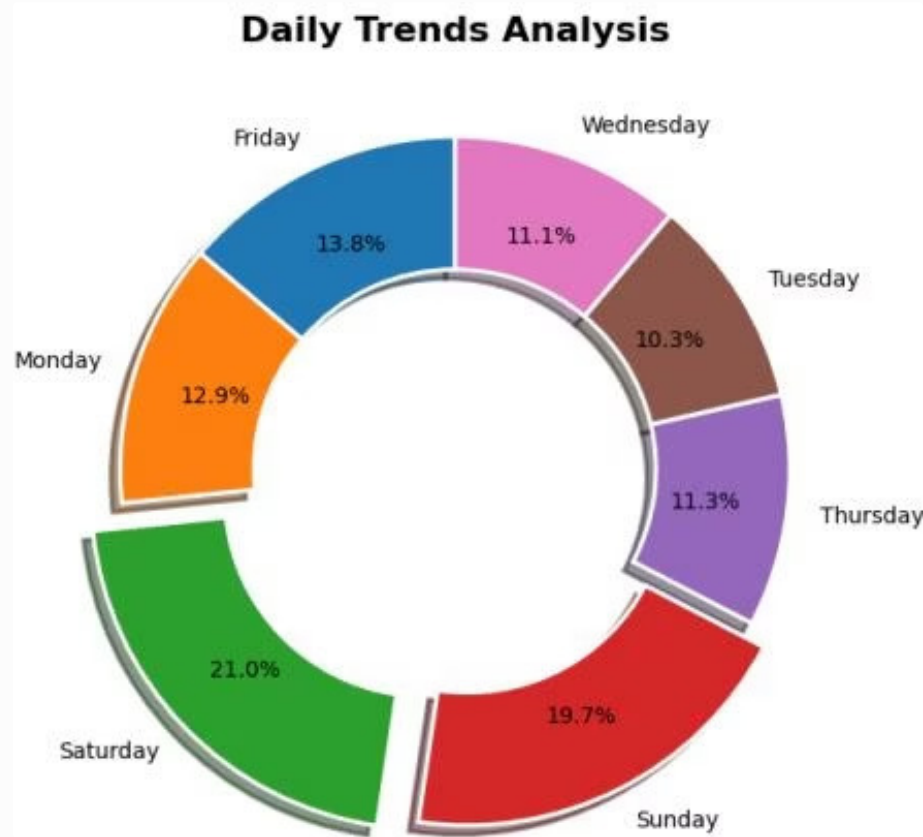
Early Morning (5–8) & Evening (16–18):
~570K & ~2.4M total sales

☀️ Peak

Afternoon (12–15): **5,376,397** —
highest volume of any period

Daily Demand Trends – Sales & Redemptions

The weekend clearly dominates: Saturday (21.0%) and Sunday (19.7%) together account for over **40% of weekly sales**. Among weekdays, Friday (13.8%) and Monday (12.9%) – days adjacent to the weekend – show comparatively higher demand. Tuesday (10.3%), Wednesday (11.1%), and Thursday (11.3%) form the off-peak core.



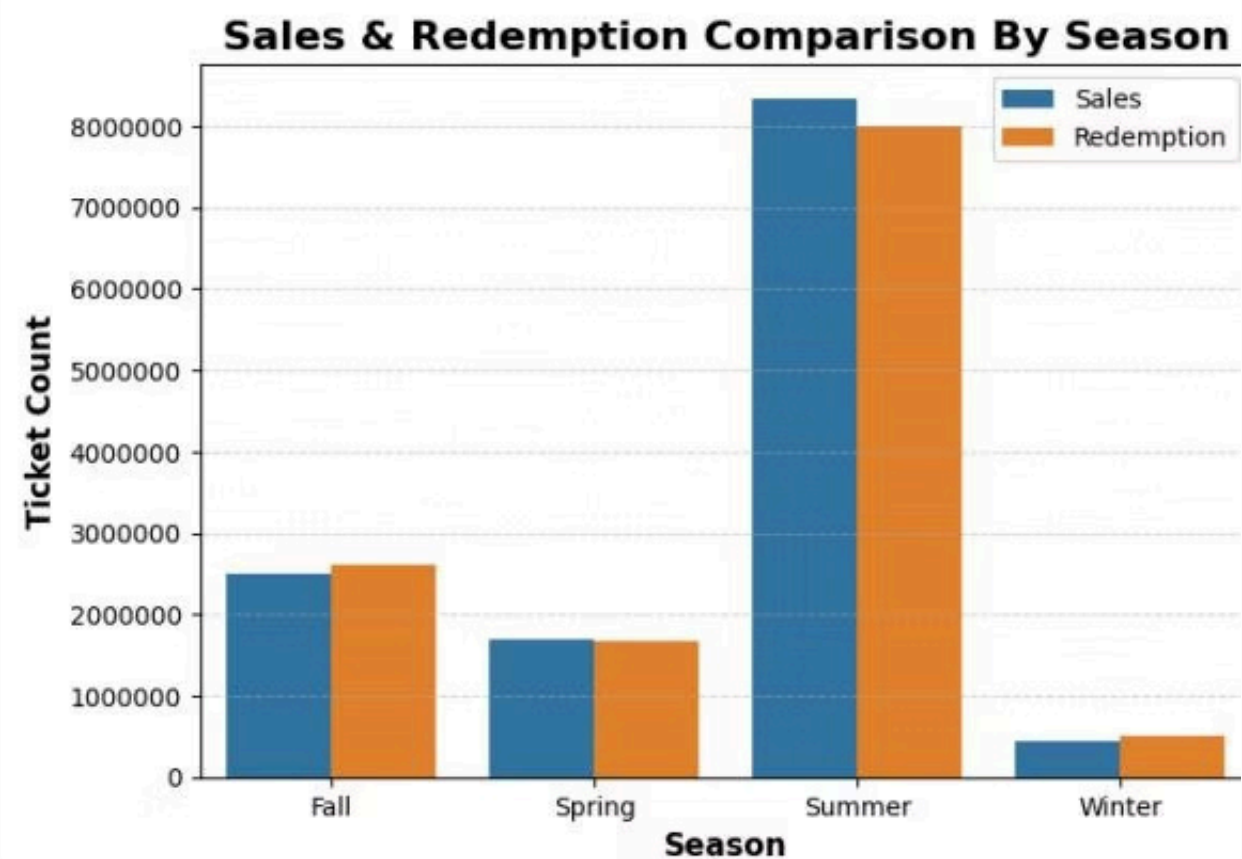
Sales vs. Redemptions by Day

Redemption patterns mirror sales almost exactly. On both weekend days, **redemption volumes slightly exceed sales**, indicating that pre-purchased tickets from adjoining days are redeemed during the weekend rush – a sign of heavy weekend footfall relative to purchase timing.

- Saturday & Sunday: highest for both metrics
- Friday & Monday: moderate weekday demand
- Tue–Thu: lowest activity – off-peak core
- Strong positive relationship: as sales rise, redemptions rise in tandem

Seasonal Comparison

Summer emerges as the peak season by a substantial margin, accounting for roughly **60% of total annual sales volume** — far outstripping the other three seasons combined. This aligns with expectations for a recreational island ferry service, where warm-weather months drive the bulk of tourism. Winter is the clear off-peak season, recording the lowest volumes; interestingly, redemptions slightly exceed sales in Winter, though totals remain very low.



8.34M

Summer Sales

Dominant peak season — ~60% of annual volume

2.50M

Fall Sales

Moderate demand, neither peak nor trough

1.69M

Spring Sales

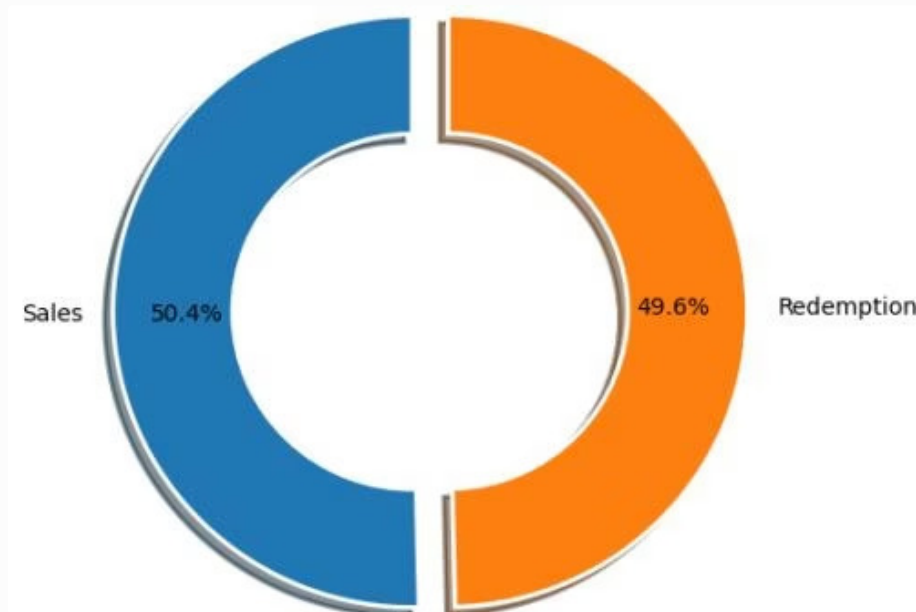
Moderate demand, similar to Fall

438K

Winter Sales

Clear off-peak — roughly 6× lower than Summer

Sales vs. Redemptions & Rolling Average Trend



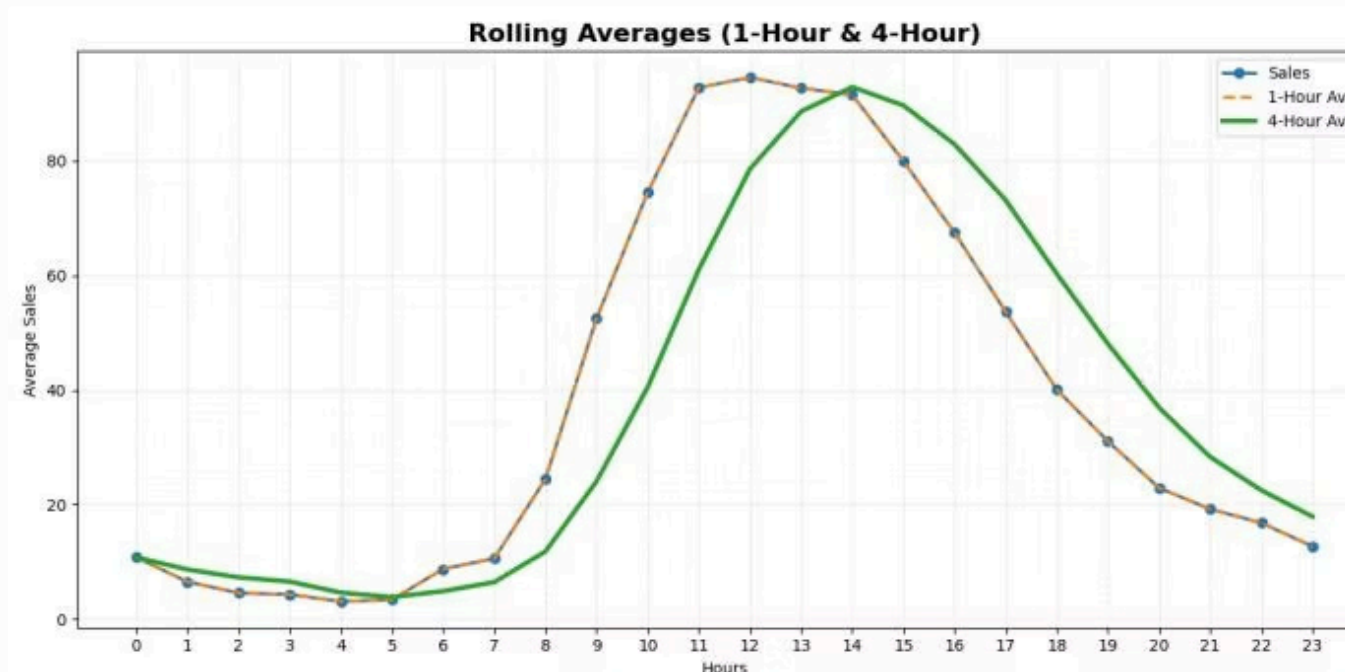
Overall Sales vs. Redemption Distribution

Ticket sales account for **50.4%** of combined volume; redemptions account for **49.6%** — a gap of only **0.8 percentage points**. This near-equal split confirms that almost all sold tickets are eventually redeemed, reflecting efficient ticket usage with minimal wastage. The redemption ratio stands at **98.56%**.



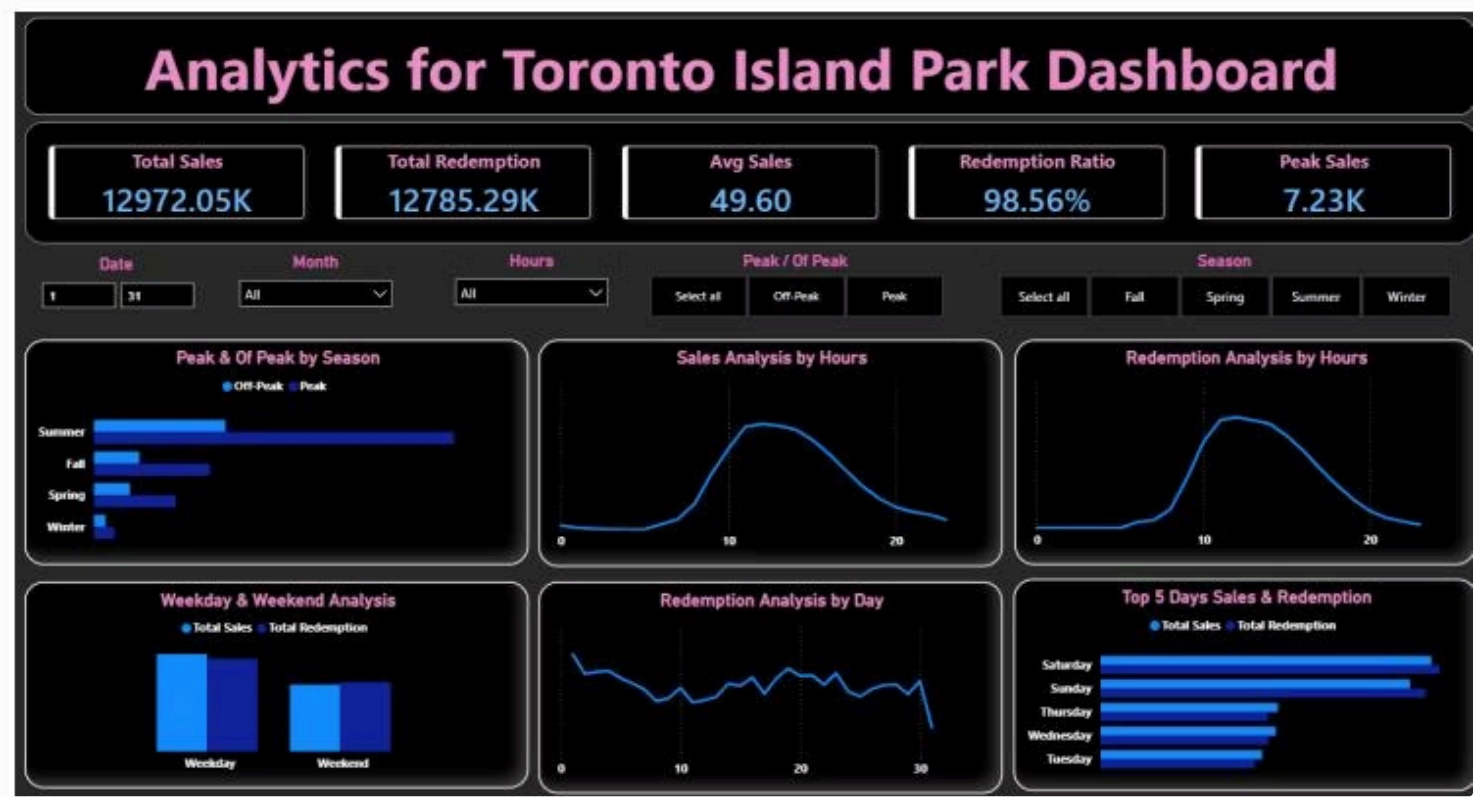
Peak sales reach 7.23K in a single interval, and the average sales per record is 49.60 — matching the descriptive statistics exactly.

Rolling Average Trend (1-Hour & 4-Hour)



The 4-hour rolling average confirms a **single, well-defined midday demand peak** between 11 AM and 2 PM, with no secondary spikes elsewhere in the day. Sales remain very low from 12 AM to 6 AM, rise rapidly after 7 AM, and gradually decline through the evening.

Interactive Dashboard & Key Findings Summary



An interactive dashboard was built atop the enriched dataset, enabling live filtering by date, month, hour, peak/off-peak status, and season. All charts and KPI cards update dynamically, visually confirming the same story as the written analysis: demand is highest in the afternoon, on weekends, and in Summer, with very few sold tickets going unredeemed.

#	Dimension	Key Finding
1	Hourly	Peak: 12–3 PM; lowest: 12 AM–5 AM
2	Daily	Sat (21.0%) + Sun (19.7%) = 40%+ of weekly sales; Tue quietest (10.3%)
3	Weekly Balance	Sales & redemptions move in lockstep; weekend redemptions slightly exceed sales
4	Seasonal	Summer ~60% of yearly sales; Winter is clear off-peak (~6× lower)
5	Sales vs. Redemption	50.4% vs. 49.6% — only 0.8-point gap; 98.56% redemption ratio
6	Trend Shape	Single midday peak confirmed; 4-hour rolling average shows no secondary spikes

Conclusions, Recommendations & Future Work

Operational Recommendations

- Scale staffing and vessel capacity during **12–3 PM**, especially on Saturdays and Sundays
- Use off-peak windows (late night, early morning, Tue–Thu) for reduced staffing, maintenance, or promotional pricing
- Anticipate a **~6× Summer surge** over Winter volumes in seasonal workforce and fleet planning
- Review ticket validity windows — weekend redemptions slightly exceed sales, risking over-commitment
- Deploy the 4-hour rolling average as a baseline for **automated demand alerts**

Limitations & Future Directions

- External factors (weather, holidays, pricing, special events) were unavailable
- Analysis is descriptive; no forecasting model (ARIMA, Prophet, ML) was built
- Fixed rule-based season/day_period bucketing; data-driven clustering could reveal more nuanced regimes
- Formal hypothesis testing (ANOVA, Kruskal-Wallis) could verify statistical significance
- Passenger/ticket-type segmentation could uncover demand drivers not visible in aggregate counts

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